

Online Appendix to “Lookout 2: Anomaly Detection via Leave-One-Out Kernel Density Estimation”

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S1 Notation and results

Section S1 restates [Proposition 1](#) and [Counterexample 1](#) of the main paper, Sections S2 and S3 prove them, and Section S4 reports additional experiments. Throughout, $\mathbf{y}_1, \dots, \mathbf{y}_n$ are sampled independently from a density f on $\mathbb{R}^m$, T is their Euclidean minimum spanning tree, which is almost surely unique, with ordered edge lengths $e_{(1)} \leq \dots \leq e_{(n-1)}$, and $h_n = e_{(\lceil \gamma(n-1) \rceil)}$ for a fixed $\gamma \in (0, 1)$. For $r > 0$, $C_n(r)$ is the number of connected components of the graph that joins every pair of observations at distance at most r . The edges of T of length at most r form a spanning forest of that graph, so there are $n - C_n(r)$ of them, and $e_{(k)} \leq r$ exactly when $C_n(r) \leq n - k$. For $\lambda > 0$, $\mathcal{P}_\lambda$ is a homogeneous Poisson process of intensity λ on $\mathbb{R}^m$; joining each pair of points

of $\mathcal{P}_\lambda \cup \{\mathbf{0}\}$ at distance at most 1, $\mathcal{C}_\lambda$ is the component containing the origin, $|\mathcal{C}_\lambda|$ is its number of points, and $\kappa(\lambda) = \mathbb{E}[1/|\mathcal{C}_\lambda|]$, with $1/|\mathcal{C}_\lambda| = 0$ when the component is infinite. Finally, b_m denotes the volume of the unit ball in $\mathbb{R}^m$, $s_{m-1} = mb_m$ the $(m-1)$ -dimensional surface measure of the unit sphere $S^{m-1} \subset \mathbb{R}^m$, and $B(\mathbf{x}, r)$ the closed ball of radius r centred at $\mathbf{x}$.

Proposition 1. *Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be sampled independently from a density f on $\mathbb{R}^m$, and fix $\gamma \in (0, 1)$. Then $n^{1/m}h_n \rightarrow q_\gamma(f)$ in probability as $n \rightarrow \infty$, where $q_\gamma(f) \in (0, \infty)$ is the unique solution q of*

$$\int_{\mathbb{R}^m} \kappa(q^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} = 1 - \gamma. \quad (\text{S1})$$

Moreover $t_- \leq q_\gamma(f) \leq t_+$, where t_- and t_+ are the unique positive solutions t of

$$\int_{\mathbb{R}^m} \exp(-b_m t^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} = 1 - \gamma \quad \text{and} \quad \int_{\mathbb{R}^m} (1 - \exp(-b_m t^m f(\mathbf{x}))) d\mathbf{x} = (1 - \gamma) b_m t^m$$

respectively, and b_m is the volume of the unit ball in $\mathbb{R}^m$.

Counterexample 1. *Let the observations be sampled independently from the density $f(\mathbf{x}) = c_m/(1 + \|\mathbf{x}\|^{m+1})$ on $\mathbb{R}^m$, $m \geq 1$, and put $\omega_n = n - \lfloor n^{1/4}/4 \rfloor$, which is at most $n-1$ once $n \geq 256$. Then $\mathbb{P}(e_{(\omega_n)} > n^{1/8}) \rightarrow 1$ as $n \rightarrow \infty$. In particular $e_{(\omega_n)}$, and every longer edge of T , tends to infinity in probability.*

S2 Proof of Proposition 1

Write $k_n = \lceil \gamma(n-1) \rceil$.

Proof of Proposition 1. Step 1: the component count. For $t > 0$ put $\Xi_n(t) := n^{-1}C_n(tn^{-1/m})$ and

$$G(t) := \int_{\mathbb{R}^m} \kappa(t^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x}.$$

Joining at distance $r_n = tn^{-1/m}$ keeps $nr_n^m = t^m$ fixed, which is the thermodynamic regime, so the law of large numbers for component counts, Mitsche & Penrose (2021, Theorem 3.13, an application of their Theorem 2.1), gives $\Xi_n(t) \rightarrow G(t)$ in probability, for every $t > 0$ and every density f . Their

limit function for component counts is κ , and their convergence is in fact complete and in L^2 . Convergence of $\Xi_n(t)$ also follows from the law of large numbers for the empirical distribution of the rescaled edge lengths of T (Penrose & Yukich 2003, Theorem 2.3(i), first obtained by Penrose 1996), since $n - C_n(r)$ is the number of edges of T of length at most r ; the component-count form is used here because its limit is expressed through κ .

Step 2: properties of G . The origin forms a component on its own exactly when $\mathcal{P}_\lambda$ has no point in $B(\mathbf{0}, 1)$, so $\kappa(\lambda) \geq \mathbb{P}(|\mathcal{C}_\lambda| = 1) = \exp(-b_m \lambda)$. Its component contains the $N \sim \text{Poisson}(b_m \lambda)$ points of $\mathcal{P}_\lambda$ in $B(\mathbf{0}, 1)$, so $\kappa(\lambda) \leq \mathbb{E}[1/(1 + N)] = (1 - \exp(-b_m \lambda))/(b_m \lambda)$. Integrating,

$$G_-(t) := \int_{\mathbb{R}^m} \exp(-b_m t^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} \leq G(t) \leq G_+(t) := \frac{1}{b_m t^m} \int_{\mathbb{R}^m} (1 - \exp(-b_m t^m f(\mathbf{x}))) d\mathbf{x}.$$

Write $G_+(t) = \int_{\mathbb{R}^m} \phi(b_m t^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x}$, where $\phi(u) = (1 - \exp(-u))/u$ for $u > 0$ and $\phi(0) = 1$, so that ϕ decreases strictly from 1 to 0 on $[0, \infty)$. By dominated convergence G_- and G_+ are continuous, tending to 1 as $t \rightarrow 0$ and to 0 as $t \rightarrow \infty$. They are strictly decreasing because their integrands decrease strictly in t wherever $f > 0$. Hence t_- and t_+ are well defined, and $G(t) \rightarrow 1$ as $t \rightarrow 0$ and $G(t) \rightarrow 0$ as $t \rightarrow \infty$.

The function κ is continuous on $(0, \infty)$, because $\lambda \mapsto \lambda \kappa(\lambda)$ is Lipschitz (Mitsche & Penrose 2021, Proposition 2.3(ii) and Section 3.6), and dominated convergence then makes G continuous. To see that G is strictly decreasing, fix $\tau > 0$, take one realisation of $\mathcal{P}_\tau \cup \{\mathbf{0}\}$, and let $\mathcal{C}(s)$ be the component of the origin when points are joined at distance s , so that $\kappa(\tau s^m) = \mathbb{E}[1/|\mathcal{C}(s)|]$. Components only grow with s , and when $s_1 < s_2$ and the nearest point to the origin lies at a distance in $(s_1, s_2]$, the origin is alone at scale s_1 and not at scale s_2 , so

$$\kappa(\tau s_1^m) - \kappa(\tau s_2^m) \geq \frac{1}{2} (\exp(-b_m \tau s_1^m) - \exp(-b_m \tau s_2^m)) > 0.$$

Hence G is strictly decreasing, because its integrand decreases strictly in t wherever $f > 0$, and (S1) has a unique solution $q_\gamma(f) \in (0, \infty)$. Since $G_-(q_\gamma(f)) \leq G(q_\gamma(f)) = 1 - \gamma = G_-(t_-)$ and G_- is decreasing, $q_\gamma(f) \geq t_-$, and the same argument with G_+ gives $q_\gamma(f) \leq t_+$.

Step 3: the quantile. Write $q = q_\gamma(f)$ and fix $\varepsilon \in (0, q)$, so that $G(q - \varepsilon) > 1 - \gamma > G(q + \varepsilon)$. Since

$h_n = e_{(k_n)}$ with $k_n = \lceil \gamma(n-1) \rceil$, $h_n \leq r$ exactly when $C_n(r) \leq n - k_n$. Putting $r = (q - \varepsilon)n^{-1/m}$ and $r = (q + \varepsilon)n^{-1/m}$ and dividing by n , $n^{1/m}h_n \leq q - \varepsilon$ exactly when $\Xi_n(q - \varepsilon) \leq 1 - k_n/n$, and $n^{1/m}h_n > q + \varepsilon$ exactly when $\Xi_n(q + \varepsilon) > 1 - k_n/n$. Since $\gamma(n-1) \leq k_n < \gamma(n-1) + 1$, we have $1 - \gamma - 1/n < 1 - k_n/n \leq 1 - \gamma + \gamma/n$, so

$$\mathbb{P}(n^{1/m}h_n \leq q - \varepsilon) \leq \mathbb{P}(\Xi_n(q - \varepsilon) \leq 1 - \gamma + \gamma/n) \quad \text{and} \quad \mathbb{P}(n^{1/m}h_n > q + \varepsilon) \leq \mathbb{P}(\Xi_n(q + \varepsilon) > 1 - \gamma - 1/n).$$

By Step 1, $\Xi_n(q - \varepsilon) \rightarrow G(q - \varepsilon) > 1 - \gamma$ and $\Xi_n(q + \varepsilon) \rightarrow G(q + \varepsilon) < 1 - \gamma$ in probability, while both thresholds tend to $1 - \gamma$. Both probabilities therefore tend to zero, so $n^{1/m}h_n \rightarrow q$ in probability. $\square$

S3 Proof of Counterexample 1

Let $f(\mathbf{x}) = c_m/(1 + \|\mathbf{x}\|^{m+1})$ on $\mathbb{R}^m$ with $m \geq 1$, where c_m is the normalizing constant, let $\mathbf{Y}$ have density f , and write $a_m = c_m s_{m-1}$. We use two elementary bounds. First, $\|\mathbf{Y}\|$ has density $a_m r^{m-1}/(1 + r^{m+1})$, which for $r \geq 1$ lies between $a_m/(2r^2)$ and a_m/r^2 , so integrating from R to ∞ ,

$$\frac{a_m}{2R} \leq p(R) := \mathbb{P}(\|\mathbf{Y}\| \geq R) \leq \frac{a_m}{R}, \quad R \geq 1. \quad (\text{S2})$$

Second, $f(\mathbf{z}) \leq c_m \|\mathbf{z}\|^{-(m+1)}$, so for every $\mathbf{x}$ and every $s \in (0, \|\mathbf{x}\|)$,

$$\mathbb{P}(\|\mathbf{Y} - \mathbf{x}\| \leq s) = \int_{B(\mathbf{x}, s)} f(\mathbf{z}) d\mathbf{z} \leq \frac{b_m c_m s^m}{(\|\mathbf{x}\| - s)^{m+1}}. \quad (\text{S3})$$

Proof of Counterexample 1. Put $R_n = a_m n^{3/4}$, $s_n = n^{1/8}$ and $K_n = n - \omega_n = \lfloor n^{1/4}/4 \rfloor$. Let N_n be the number of observations with $\|\mathbf{y}_i\| \geq R_n$, and let M_n be the number of pairs $\{i, j\}$ with $\|\mathbf{y}_i\| \geq R_n - s_n$, $\|\mathbf{y}_j\| \geq R_n - s_n$ and $\|\mathbf{y}_i - \mathbf{y}_j\| \leq s_n$. Throughout, $n \geq 256$ and n is large enough that $R_n - 2s_n \geq \max(1, R_n/2)$.

Step 1: the number of distant observations. N_n is binomial with n trials and success probability $p(R_n)$, which by (S2) lies in $[n^{-3/4}/2, n^{-3/4}]$. Its mean μ_n therefore satisfies $n^{1/4}/2 \leq \mu_n \leq n^{1/4}$, its variance is at most μ_n , and $K_n \leq \mu_n/2$. By Chebyshev's inequality,

$$\mathbb{P}(N_n \leq K_n) \leq \mathbb{P}(|N_n - \mu_n| \geq \mu_n/2) \leq \frac{4}{\mu_n} \leq 8n^{-1/4} \rightarrow 0.$$

Step 2: no close pairs among distant observations. Conditioning on the position of the first point of a pair and applying (S2) and (S3),

$$\mathbb{E}[M_n] \leq \binom{n}{2} p(R_n - s_n) \sup_{\|\mathbf{x}\| \geq R_n - s_n} \mathbb{P}(\|\mathbf{Y} - \mathbf{x}\| \leq s_n) \leq \frac{n^2}{2} \cdot \frac{a_m}{R_n - s_n} \cdot \frac{b_m c_m s_n^m}{(R_n - 2s_n)^{m+1}} \leq \frac{2^{m+1} a_m b_m c_m n^2 s_n^m}{R_n^{m+2}}.$$

The right side is $C_m n^{2+m/8-3(m+2)/4} = C_m n^{1/2-5m/8}$ with C_m depending only on m , and tends to zero for every $m \geq 1$. Hence $\mathbb{P}(M_n \geq 1) \leq \mathbb{E}[M_n] \rightarrow 0$.

Step 3: conclusion. Suppose $N_n > K_n$ and $M_n = 0$. Any observation within distance s_n of an observation $\mathbf{y}_i$ with $\|\mathbf{y}_i\| \geq R_n$ has norm at least $R_n - s_n$, so the pair would be counted in M_n ; since $M_n = 0$, there is no such observation. Each of the N_n observations beyond R_n is therefore a component on its own when observations are joined at distance s_n , so $C_n(s_n) \geq N_n > K_n = n - \omega_n$. Exactly $n - C_n(s_n)$ edges of T have length at most s_n , which is fewer than ω_n , so $e_{(\omega_n)} > s_n$. Therefore

$$\mathbb{P}(e_{(\omega_n)} > n^{1/8}) \geq 1 - \mathbb{P}(N_n \leq K_n) - \mathbb{P}(M_n \geq 1) \rightarrow 1. \quad \square$$

S4 Additional experiments

This section gives two further comparisons of the original and new versions of lookout, a set of showcase examples, and the details of the bandwidth sensitivity study of Section 4.6 of the main paper. Both versions use the default parameters $\alpha = 0.01$, $\beta = 0.9$ and $\gamma = 0.98$, with scaling applied.

S4.1 Experiment S1: Different anomaly means for Normal distribution

The non-anomalous observations have independent $\mathcal{N}(0, 1)$ coordinates in $\mathbb{R}^2$, and the anomalies independent $\mathcal{N}(\mu, 1)$ coordinates with $\mu \in \{2.50, 2.75, \dots, 4.00\}$. Each replication has 1000 non-anomalies and 10 anomalies, and there are 20 replications for each value of μ . Fig. S1 shows the data and results. As μ increases the anomalies move away from the non-anomalies and the true

positive rate rises. The two versions have similar true positive rates at every value of μ . The false positive rate is about 0.008 for the new version and between 0.002 and 0.005 for the original.

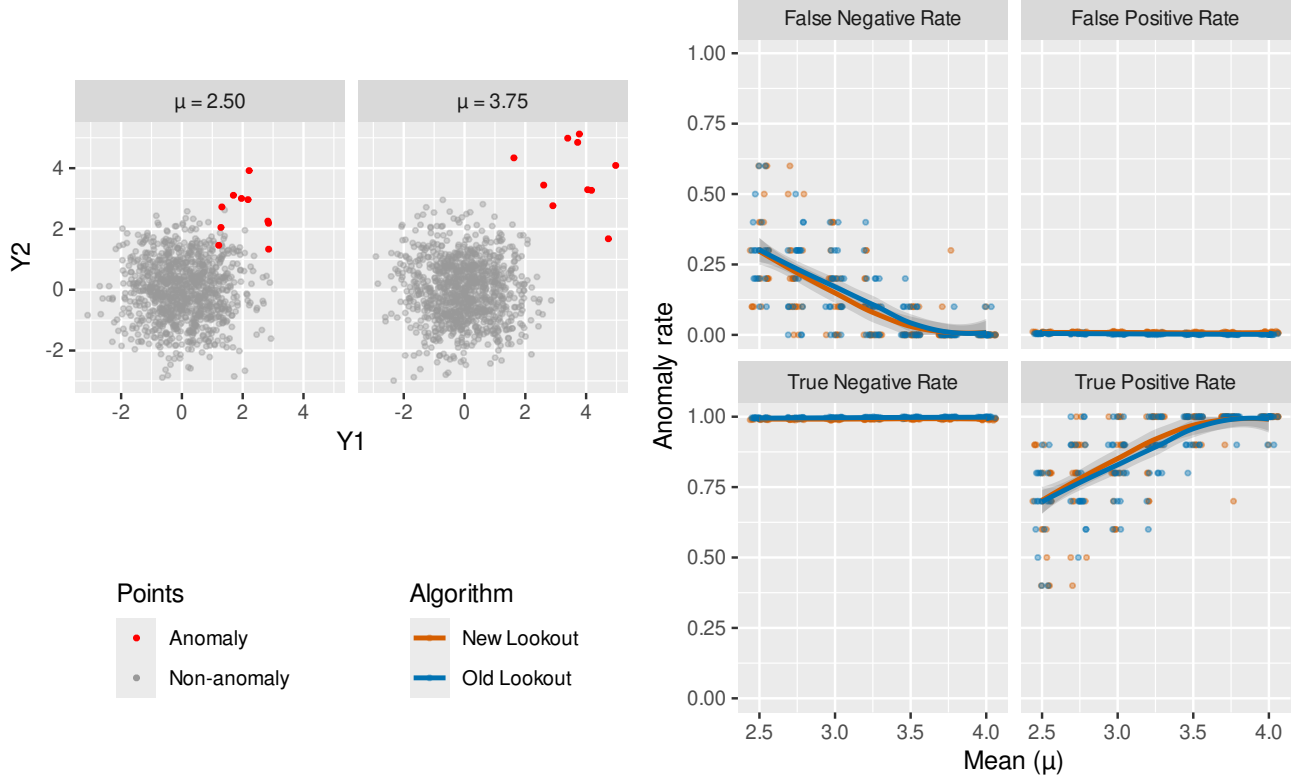


Figure S1: *Experiment S1. Left: data with $\mu = 2.5$ and $\mu = 3.75$; anomalies with higher values of μ are easier to identify. Right: error rates of the two versions of lookout, with points jittered horizontally.*

S4.2 Experiment S2: Anomalies on the boundary as n increases, Gamma distribution

This experiment repeats Experiment 2 of the main paper with gamma-distributed observations. The n non-anomalous observations have independent gamma coordinates in $\mathbb{R}^2$ with shape 2 and rate 2. For the anomalies, we generate n further points with independent gamma coordinates with shape 2.2 and rate 2, keep those whose distance from the origin exceeds the 0.99 quantile of the distances of the non-anomalous observations, and sample 0.005 n of them. There are 10 replications

for each $n \in \{1000, 2000, \dots, 10000\}$, and the kernel density estimates use the truncated sum with 100 neighbours. Fig. S2 shows the data for $n = 10000$ and the results. The true positive rate of the new version is between 0.76 and 0.84, and at least as high as that of the original at every n . The false positive rate is about 0.012 for the new version and 0.008 for the original.



Figure S2: *Experiment S2. Left: data for $n = 10000$. Right: error rates of the two versions of lookout, with points jittered horizontally.*

S4.3 Showcase examples

Fig. S3 applies both versions to five examples in $\mathbb{R}^2$, to check that the new version agrees with the original on obvious anomalies. In the first row, five anomalies lie at the top right of a standard normal sample. In the second, five anomalies lie midway between two normal clusters. In the third, a group of three anomalies lies away from three normal clusters, and in the fourth the three anomalies are spread out. In each of these four rows both versions identify all the anomalies, and

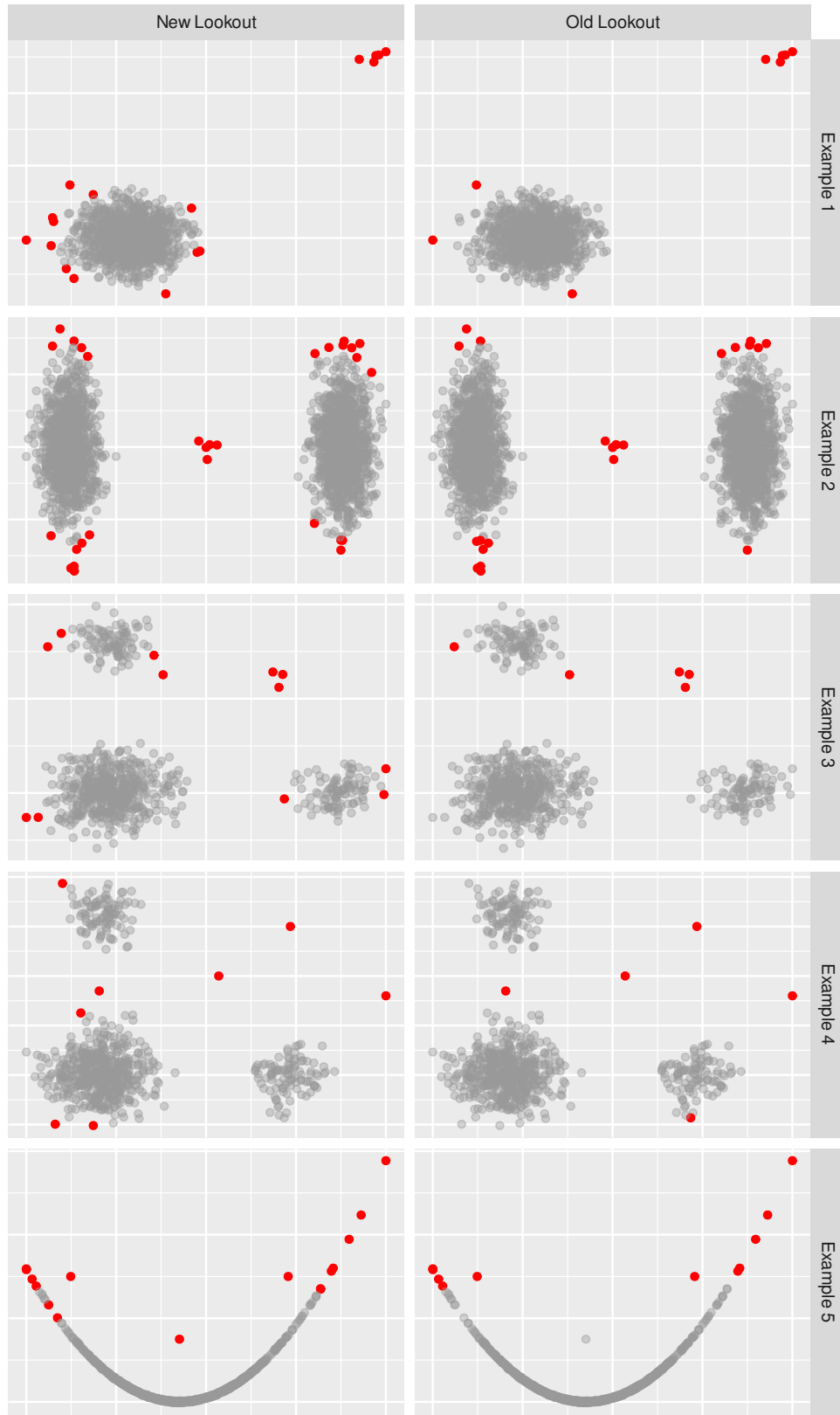


Figure S3: *Showcase examples. Anomalies identified by the new (left) and original (right) versions of lookout are in red.*

the new version flags more non-anomalous points: 12, 24, 9 and 5 against 3, 17, 2 and 2 for the original. In the last row the non-anomalous points lie on a parabola and three anomalies lie off it. Neither version takes account of the geometry of the curve, so this example is difficult. The new version identifies all three anomalies and the original two, and they flag 13 and 9 points on the curve. Overall, the new version is more sensitive to anomalies and produces more false positives. Its false positives tend to be in regions of low density, which is consistent with our definition of an anomaly.

S4.4 Sensitivity to the bandwidth

Section 2.3 of the main paper argues that the bandwidth should be of the order of the spacing in the tails, and that neither the constant nor MISE-optimality matters much. To check this, we repeat Experiments 1, 2 and 3 of the main paper, and Experiment S1, with the bandwidth h_n multiplied by 1/4, 1/2, 2 and 4, and with h_n replaced by the bandwidth h_{opt} of Table 1 of the main paper, the AMISE-optimal bandwidth for the Epanechnikov kernel and a standard normal density, computed for the standardized data. Everything else in Algorithm 2 of the main paper is unchanged. In Experiment 2 we use $n \in \{1000, 2000, 4000\}$ and the exact leave-one-out estimate, so that the truncated sum of Section 3 of the main paper does not affect the results at large multiples of h_n . Each setting is replicated ten times. [Fig. S4](#) reports the area under the ROC curve computed from the anomaly scores, and the true and false positive rates at $\alpha = 0.01$.

The ranking of the observations is insensitive to the bandwidth. The area under the ROC curve is almost the same for every bandwidth from $h_n/2$ to $4h_n$ in all four experiments, apart from three iterations of Experiment 3 at $4h_n$. At $h_n/4$ it is lower in Experiments 2 and S1, and in Experiment 3 from the fourth iteration. The threshold is more sensitive. At h_n the false positive rate is between 0.005 and 0.013 in the four experiments, close to the nominal $\alpha = 0.01$. At $h_n/4$ it is between 0.05 and 0.27, because observations with no other observation inside the kernel support have a leave-one-out estimate of zero and are all declared anomalous; in Experiment 3 the GPD

could not be fitted in 10 of the 100 runs at $h_n/4$. Doubling or quadrupling the bandwidth keeps the false positive rate at or below the nominal level and, in Experiments 1 and 2, lowers the true positive rate, because the estimate at an isolated observation then includes contributions from the bulk of the sample. In Experiments 2 and S1, with normal observations in two dimensions, h_{opt} is on average about h_n and $0.75h_n$ respectively, and gives similar results. In Experiment 1, with gamma observations, it is about a third of h_n and gives a false positive rate of 0.02. In Experiment 3, where $m = 6$, it is about a quarter of h_n ; the GPD could not be fitted in 73 of the 100 runs, because more than 10% of the observations had no other observation inside the kernel support, and in the remaining runs the false positive rate averaged 0.34. As described in Section 2.3 of the main paper, in moderate dimensions a bandwidth chosen for the bulk of the distribution is too small for the tails.

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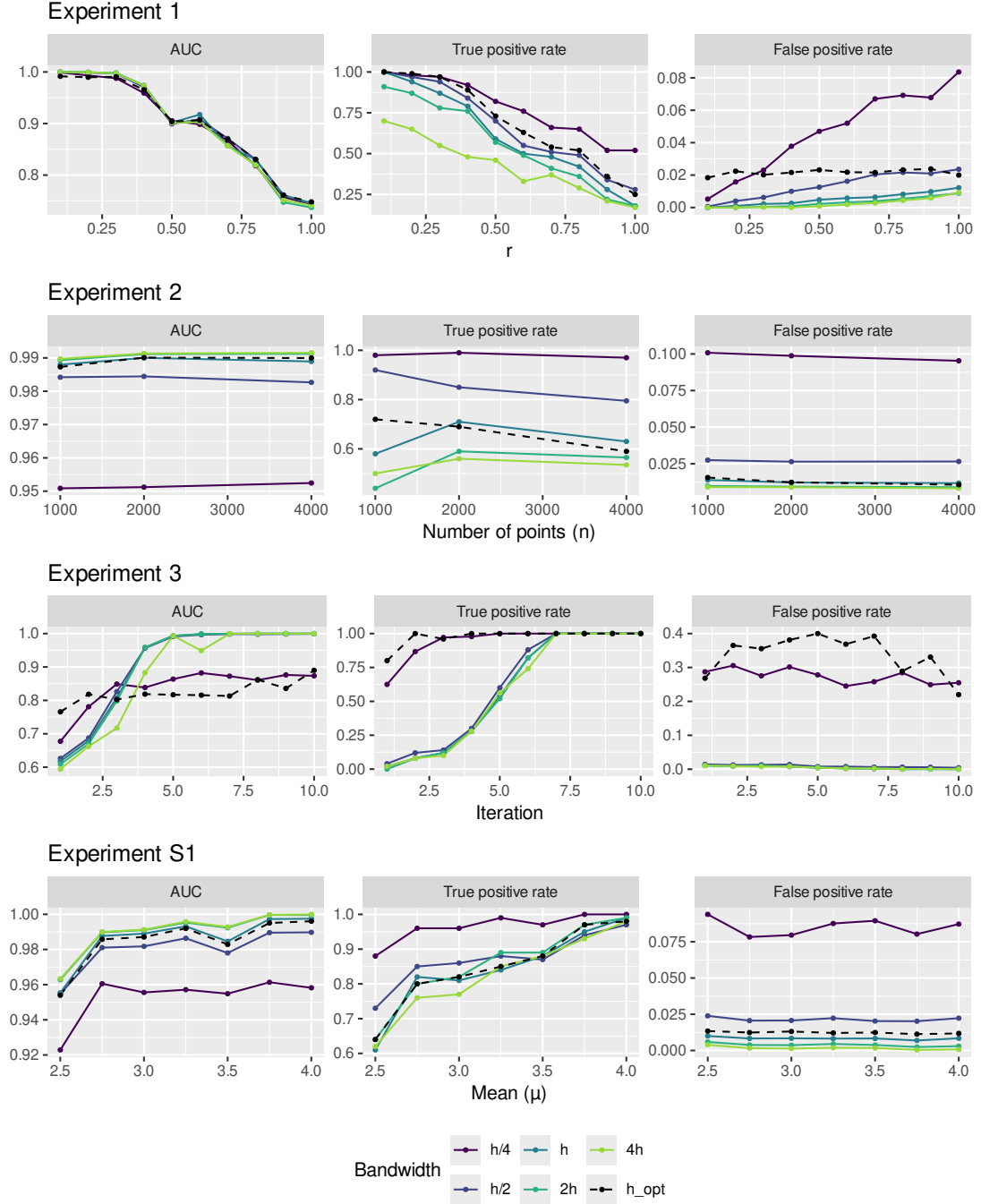


Figure S4: Sensitivity of the new version of lookout to the bandwidth in Experiments 1, 2 and 3 of the main paper and Experiment S1: mean over ten replications of the area under the ROC curve and of the true and false positive rates, with the minimum spanning tree bandwidth h_n multiplied by 1/4, 1/2, 1, 2 and 4, and with the AMISE-optimal bandwidth h_{opt} of Table 1 of the main paper (dashed).